

Supplementary Material: Beyond Static Imitation: Auditing LLM-Derived Traveler Responses for Transport Simulation

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ABSTRACT

This supplement documents the traveler inputs and stated-choice tasks, the specification and training of the Student models, the complete comparisons with stated choices and with the contemporary Teacher, the simulation experiments and the computational measurements. It follows the order of the three questions in the main article, which concern fidelity to Teacher responses, agreement with stated responses and preservation of responses in simulation.

S1. Traveler inputs and stated-choice tasks

S1.1. Traveler state and public-transport supply

Each traveler state combines nine categorical fields and 17 global numerical fields with 12 numerical attributes for every alternative. The categorical fields describe age, income, occupation, habitual mode, schedule flexibility, mobility limitation, trip purpose, time constraint and weather. The global fields cover household composition, access to a car, licence, bicycle and transit pass, trip distance, desired departure and arrival times, and contextual conditions. Alternative attributes include travel time, monetary cost, access time, transfers, reliability delay and weather exposure. Table S1 defines the additional PT components and the feasibility information. PT accessibility is derived from a

Table S1

Public transport supply attributes used to condition the Student model.

Attribute	Behavioral role
Access walking time	Burden from origin to boarding stop
Egress walking time	Burden from alighting stop to destination
Waiting time	Schedule-dependent waiting for a connection
In-vehicle time	Time aboard public transport
Transfer time	Time associated with transferring between services
Coverage score	In-vehicle distance divided by OD distance, clipped to [0, 1]
Feasibility flag	Whether the specified search identifies a PT connection

route search over direct and one-transfer connections, with a minimum transfer time of 180 seconds and an extended egress search of up to 1.5 km. Coverage is the in-vehicle distance divided by the origin–destination distance, clipped to [0, 1]. Accessibility classes are assigned in sequence. Class A requires access and egress of at most five minutes each, coverage of at least 0.85, at most one transfer and a door-to-door time of at most 45 minutes. Class B allows eight minutes of access or egress and coverage of at least 0.7 with one transfer. Class C allows twelve minutes of access or egress, coverage of at least 0.5 and at most two transfers. Class D contains the remaining feasible routes, and class E denotes that no connection was found. Although class C admits two transfers, the search itself returns at most one. Feasibility is therefore defined by the search, and an unsuccessful search does not rule out every physically possible connection.

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S1.2. Questionnaire conditions and model encodings

The Singapore questionnaire shows respondents numerical tables of time and cost (Table S2). The Shanghai questionnaire describes conditions qualitatively and states only three values explicitly, a fare of CNY 6, a parking cost of CNY 30 and one additional transfer. The remaining times and their division into PT components are reference values used for model evaluation (Table S3) and were not shown to respondents. The two surveys therefore specify their tasks with different numerical precision. The questionnaire's car option is represented by the private-car alternative in

Table S2

Displayed task attributes used in the four-mode analysis. Each column contains a baseline/intervention pair; arrows identify the changed numeric attribute. Times are minutes and costs are SGD. Other numeric attributes remain unchanged within the pair.

Pair (tasks base/int.)	A (1/6)	W (3/7)	D (5/9)	F (8/2)	R (10/4)
Distance (km)	4.2	2.5	8.5	5.3	6.0
Car time	10	8.5	18	12	24 → 34
PT time	32 → 48	26	38 → 53	32	32
Bicycle time	20	12.5	45	24	28
Walking time	65	38	140	75	80
Private-car cost	7.50	6.50	10.10	8.20	9.00
PT fare	2.80	2.40	3.30	2.80 → 4.20	2.20
PT access walk	3 → 12	4	5	4	4
PT transfers	0 → 2	0	1	0	0

Bicycle and walking costs are zero. W changes clear weather to heavy rain: car and PT are described as sheltered, bicycle and walk as exposed. R describes local roadworks, lane restrictions, and a detour. D describes a sudden PT delay. One choice is scored per task, and the car option is represented by the private-car alternative.

Table S3

Shanghai scenario wording and the numerical reference encoding used for model evaluation. Numerical reference values were not displayed as a task table in the deployed questionnaire.

Task	Condition stated to respondents	Additional numerical reference encoding
B0	Clear weather; normal traffic	Times car/PT/bicycle/walk: 20/35/25/80 min; car cost 15 yuan, PT fare 4 yuan.
W1	Rain; normal traffic	Rain intensity 0.75; travel times unchanged.
D1	Clear weather; PT delayed	PT time 35 → 50 min, waiting 5 → 20 min; context delay 15 min.
WD1	Rain and PT delay	W1 and D1 reference changes combined.
F1	Clear weather; PT fare rises to 6 yuan	Fare 4 → 6 yuan; baseline 4 yuan is imputed.
P1	Clear weather; parking fee rises to 30 yuan	Total car cost 15 → 35 yuan: parking 10 → 30 plus 5 yuan imputed other cost; parking multiplier 3.
R1	Clear weather; road temporarily obstructed	Car time 20 → 40 min and road-disruption flag.
A_WALK	Clear weather; PT stops farther away	Access and egress each 7 → 12 min; PT total 45 min.
A_WAIT	Clear weather; fewer PT services	Waiting 5 → 15 min; PT total 45 min.
A_TRANSFER	Clear weather; one PT transfer	One transfer and 10 min transfer time; PT total 45 min.

The analysis uses private car, public transport, conventional bicycle and walking. Answers indicating that none of the offered modes is suitable have no four-mode label. A common reference trip has distance 6 km, desired departure 08:00 and desired arrival 09:00. Baseline PT components are access/egress/wait/in-vehicle/transfer times 7/7/5/16/0 min, coverage 0.8 and a feasible-PT flag. Household size 2 and unmeasured numerical attributes are reference values, not measurements of respondents' trips. Only the fare of 6 yuan, the parking fee of 30 yuan and one transfer are stated numerically in the questionnaire.

the four-mode analysis, which leaves the recorded choices and the evaluation denominators unchanged. The model makes private car available only to respondents who own a car and hold a driving licence, and bicycle only to those with bicycle access. Every explicit choice is scored, including eight Singapore car choices and two Shanghai bicycle choices made by respondents for whom the model treats that mode as unavailable. Because NLL is infinite for these choices, reported NLL uses a probability floor of $\epsilon = 10^{-8}$, and the number of such choices is reported separately.

S1.3. Sample composition and collection period

In Singapore, 334 respondents gave consent and completed the survey. Two visitors were screened out, leaving 332 residents and 3,320 explicit choices. In Shanghai, 341 respondents gave consent, and eight who had not travelled in Shanghai during the preceding 30 days left at screening. All 333 eligible respondents completed the ten tasks. The model's input mapping covers 321 of them. Eleven others reported habitual-mode or departure-schedule categories that the model does not represent, and one declined to report income. Their answers are retained for the human-only sensitivity analysis below.

The 321 Shanghai respondents made 3,208 explicit mode choices and gave two answers indicating that none of the modes was suitable. These two answers are not scored in the four-mode analysis. Contrast denominators are 321 for rain, delay, fare, parking, road and transfer-versus-waiting, and 320 for the interaction and walking-versus-waiting contrasts. Every Singapore contrast has 332 respondents. In Singapore, 23 missing or undisclosed incomes are assigned the medium income category, with low and high incomes examined as sensitivity settings. Responses were submitted from 20 August to 5 September 2026 in Singapore and from 12 to 18 September 2026 in Shanghai, and both questionnaires recorded participant consent.

The electronic questionnaires were circulated through WeChat, WhatsApp and personal networks, including onward sharing by family members, friends, classmates and colleagues. Recruitment was therefore by convenience and snowball sampling. The Singapore questionnaire was administered in Google Forms and the Shanghai questionnaire in Wenjuanxing. The instruments are identified by the [Singapore form](#) (owner/editor access may be required) and the [Shanghai questionnaire](#). These instrument links do not grant access to individual responses. A privacy statement preceded participation, both instruments recorded consent, and the study was conducted on an ethics-exemption basis. The questionnaires did not solicit sensitive personal information.

S1.4. Reference values and their sensitivity

Attributes that were not collected take fixed reference values. Singapore uses a household size of two, medium schedule flexibility, no transit pass and a reference departure at 08:00. Shanghai uses a household size of two and the numerical reference encoding in Table S3. The twelve Singapore sensitivity profiles vary car costs, the mapping of missing income and occupation, transit pass, flexibility, purpose, departure time, PT components and coverage, the allocation of delay and rain intensity. The eight Shanghai profiles vary household size, coverage, imputed durations and the allocation of three minutes among PT components while total PT time is held fixed.

All profiles apply the model's availability rule to all 14 fitted models. Table S4 reports the minimum and maximum across these specifications, which are ranges rather than confidence intervals. The profiles were defined without reference to agreement with human choices. They cover selected reference values and time allocations but not every plausible reading of the qualitative scenarios, and in particular they do not vary the intensity implied by the Shanghai delay description. The Shanghai coverage analysis compares the 321 respondents covered by the model with all 333 eligible respondents who completed the survey. A nested respondent bootstrap with 10,000 draws (seed 917) resamples the 333 people and computes both cohort responses in each draw, preserving their overlap and all ten answers. The twelve additional respondents contribute 120 choices, of which 27 are car, 88 PT, none bicycle and five walking. No human response changes by more than 0.26 percentage points, and all eight point estimates keep their sign (Table S5).

Table S4

Sensitivity to reference values under the model availability rule. Accuracy is in percent; PT-share and response biases are model minus human in percentage points. Ranges cover all specified profiles.

Model	Accuracy	PT-share bias	Delay bias	Access-contrast bias
<i>Singapore (12 profiles)</i>				
SA-Student	[66.0, 68.2]	[−31.8, −28.3]	[−4.6, −0.7]	[9.4, 11.4]
MNL-S	[70.4, 72.2]	[−9.4, 1.0]	[3.1, 3.5]	[37.7, 40.2]
Soft KL	[71.3, 73.1]	[−19.5, −15.0]	[−3.4, 3.8]	[13.0, 17.9]
CE+KL	[67.7, 71.5]	[−25.6, −19.7]	[3.0, 8.8]	[−0.6, 4.8]
Signed response	[72.1, 73.6]	[−18.3, −13.6]	[−3.3, 3.9]	[15.3, 20.5]
Direction+magnitude	[70.8, 72.2]	[−19.0, −14.2]	[−4.6, 2.5]	[13.3, 18.3]
<i>Shanghai (eight profiles)</i>				
SA-Student	[74.9, 81.3]	[−22.2, −14.6]	[−16.5, −10.2]	[−13.7, −10.6]
MNL-S	[80.5, 82.6]	[−9.4, −2.9]	[4.7, 7.6]	[−9.7, −7.4]
Soft KL	[76.7, 80.5]	[−14.7, −10.2]	[−5.3, −3.1]	[−11.1, −9.2]
CE+KL	[75.0, 77.4]	[−17.3, −13.5]	[−7.6, −1.5]	[−12.2, −9.1]
Signed response	[77.1, 80.5]	[−14.3, −8.5]	[−3.6, −1.0]	[−12.1, −10.1]
Direction+magnitude	[77.9, 81.2]	[−13.5, −8.6]	[−3.9, −0.7]	[−11.1, −9.3]

The access contrast is poorer PT accessibility in Singapore and walking versus waiting in Shanghai; these are different task definitions. Singapore has 3,320 scored choices and 332 complete respondents per contrast. Shanghai has 3,208 choices, 321 complete delay respondents and 320 for walking versus waiting. The eight Singapore and two Shanghai choices of modes the model treats as unavailable remain scored. Neural accuracy is the mean of individual-seed accuracies, not classification from averaged probabilities.

Table S5

Shanghai human responses with and without the restriction to respondents covered by the model input mapping. Changes and differences are percentage points; the last column is complete eligible respondents minus the model-evaluated cohort with a nominal 95% nested respondent-bootstrap interval.

Contrast	N_{321}	N_{333}	Mapped	Complete	Difference [95% interval]
Rain	321	333	4.67	4.50	-0.17 [-0.39, 0.00]
PT delay	321	333	-14.64	-14.41	+0.23 [-0.45, 0.75]
Rain × delay	320	332	9.38	9.34	-0.04 [-0.53, 0.66]
PT fare	321	333	-5.92	-6.01	-0.09 [-0.78, 0.37]
Parking	321	333	-9.35	-9.31	+0.04 [-0.63, 0.48]
Road penalty	321	333	-9.35	-9.61	-0.26 [-1.16, 0.37]
Walk versus wait	320	332	1.88	1.81	-0.07 [-0.25, 0.09]
Transfer versus wait	321	333	2.49	2.70	+0.21 [-0.21, 0.92]

PT is the outcome except parking and road disruption, which use the car response. All additional 12 respondents contribute explicit answers to every contrast. The two none-suitable responses reduce the interaction and walk-versus-wait counts from 321/333 to 320/332. These are complete-case denominators and do not exclude further respondents. Bonferroni-adjusted intervals and intervals for each cohort are given in the supplementary data files.

Table S6 lists the encoded fields.

Table S6

Fields of the encoded traveler state. Mode availability is represented separately from these attributes.

Input group	Fields
Categorical traveler fields	age group, income group, occupation, habitual mode, schedule flexibility, mobility limitation
Categorical trip/context fields	trip purpose, time constraint, weather condition
Numerical traveler fields	household size, presence of children, car ownership, driving licence, bicycle ownership, transit pass
Numerical trip fields	distance (km), desired departure time, desired arrival time
Numerical context fields	weather intensity, road congestion, PT delay, PT disruption, road disruption, fare multiplier, parking cost multiplier, congestion charge
Alternative fields	travel time, monetary cost, access time, number of transfers, reliability delay, weather exposure
Additional PT-supply fields	PT feasibility, egress time, waiting time, in-vehicle time, transfer time, coverage ratio

The nine categorical and seventeen global numerical fields form the shared state; each mode has twelve numerical attributes. Times are in minutes and distance in kilometres before encoding. Categorical vocabularies and numerical normalization are fitted on training data. For SA-Student, existing statistics are inherited and only the six added supply fields receive newly fitted normalization. The supplementary data files give the variable names used in the implementation.

S2. Controlled learning and departure prediction

S2.1. Training and model selection

The controlled benchmark contains 1,839 training, 365 validation and 437 test states from disjoint groups of 28, six and six personas. The test set comprises 226 single-context, 51 accessibility, 96 joint-context and 64 mechanism states, which define 398 response pairs and 94 interaction contrasts. Training proceeds over pairs, and the 1,681 training pairs expose the model to 3,362 states per epoch. Within each training seed (42, 2026 and 7), the four neural objectives share initialization, feature encoding, the order of training pairs and a budget of 6,360 updates over 120 epochs. They also share the availability rule, under which a mode enters the response loss if it is available in either state of a pair.

Every neural Student has 24,562 parameters. The nine categorical fields receive eight-dimensional embeddings and are combined with 17 standardized global features in a two-layer encoder of width 64. A mode embedding and the 12 alternative attributes enter a two-layer encoder of width 32, and a scorer with 64 hidden units maps both representations to utilities. The departure predictor uses the global representation together with the mean representation of the available alternatives. Hidden layers use ReLU activations, and the global encoder and scorer apply a dropout rate of 0.1. All models are trained with Adam at a learning rate of 1.25×10^{-4} and weight decay of 10^{-4} , in batches of 32 pairs, with a Huber loss on departure adjustment whose transition is one minute. For the probability-response comparison, the epoch with the lowest validation macro-source KL is retained.

MNL-S specifies four linear utilities over 111 explanatory variables. It minimizes cross-entropy against the soft Teacher targets with an L_2 penalty, whose weight of 10^{-4} was chosen from $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ by validation macro-source KL. Estimation by L-BFGS-B converges within 2,000 iterations. The ridge departure predictor uses a penalty of 10^{-3} , chosen by validation departure MAE. Table S8 reports feasibility and departure-tail metrics for the models selected on choice.

Response and feasibility metrics. For response pair i , let $\mathcal{A}_i$ be the set of alternatives available in either of its two states. Probability-response error is

$$E_{\Delta p} = \frac{1}{N_{\text{pair}}} \sum_{i=1}^{N_{\text{pair}}} \frac{1}{|\mathcal{A}_i|} \sum_{m \in \mathcal{A}_i} |\Delta p_{im}^S - \Delta p_{im}^T|. \quad (\text{S1})$$

Fidelity at individual states is measured by $D_{\text{KL}}(\mathbf{p}_T \| \mathbf{p}_S) = \sum_{m: p_{T,m} > 0} p_{T,m} \log(p_{T,m}/p_{S,m})$, and macro-source KL summarizes this divergence at the level of the four supervision sources. For a linked baseline 0, individual perturbations a , b and their combination ab , the mode-specific interaction is

$$I_{im}^j = p_{im}^j(ab) - p_{im}^j(a) - p_{im}^j(b) + p_{im}^j(0), \quad j \in \{T, S\}. \quad (\text{S2})$$

Table S7

Construction of the four behavioral-supervision sources and their contributions to the controlled test set.

Source	Comparison represented	States	Pairs
Single-context	A traveler–trip baseline and single-axis changes in weather, congestion, PT delay, fares, parking or road disruption.	226	214
Accessibility	The same traveler and trip under PT supply profiles, varying access, waiting, transfers and connection feasibility.	51	40
Joint context	Two perturbations applied together, linked to their baseline and single-perturbation counterparts where available.	96	96
Mechanism	A baseline and three context–attribute variants: both changed, context only, or attributes only. These separate different encoded paths of the same intervention.	64	48
Total	Held-out persona groups across the four sources.	437	398

A state combines a traveler, a trip, a context, mode attributes and Teacher targets. A response pair links two states, and a state can belong to several pairs. Persona groups are disjoint across training, validation and test (28/6/6). Accessibility data additionally use disjoint OD pools. Teacher probability vectors are averaged over valid repeated queries. The counts are states and pairs, not independent travelers.

Interaction error averages $|I_{im}^S - I_{im}^T|$ over the modes available in a linked contrast and then over the 94 evaluated contrasts. It therefore measures how two perturbations combine, rather than the response to a single intervention. With $\mathcal{E}$ denoting the test states in which no PT route is found, the feasibility violation rate is

$$\text{FVR} = \frac{1}{|\mathcal{E}|} \sum_{x \in \mathcal{E}} \mathbf{1} \left\{ \arg \max_{m \in \mathcal{A}(x)} p_{S,m}(x) = \text{PT} \right\}. \quad (\text{S3})$$

The controlled evaluation contains $|\mathcal{E}| = 12$ such states. FVR counts how often PT is the most probable mode where the route search finds no connection, which differs from the mean PT probability on those states. Departure MAE averages $|\tau_S - \tau_T|$ over the evaluated states.

Table S8

Feasibility and departure diagnostics on the same controlled test pool. Neural entries average three training seeds. FVR uses 12 infeasible states per seed. Departure errors are in minutes; response SD describes training-seed variation.

Model	FVR	Departure median	Departure P90	Response SD
Soft KL	0.0833	5.12	15.59	0.0032
CE + KL	0.0833	5.16	15.74	0.0016
Signed response	0.0278	5.18	15.37	0.0050
Direction + magnitude	0.0556	5.06	16.06	0.0042
MNL-S	0.0000	10.53	25.42	–

S2.2. Independent departure predictors

The modular comparison keeps the 444 MNL-S choice coefficients fixed and adds either a bounded linear predictor with 111 parameters or an independent neural predictor with 18,289 parameters. The neural predictor keeps the feature encoders and departure layer of the joint architecture, omits the utility scorer and is trained on the departure Huber loss alone. The two modular Students therefore have 555 and 18,733 parameters in total.

All timing comparisons share the training states, the three seeds, the optimizer and the budget of 6,360 updates. The retained epoch minimizes validation departure MAE, with ties resolved in favour of the earliest epoch, and the four joint objectives are retrained under the same criterion. These timing fits are separate from the fits used for the probability-response comparison, although both follow the same update budget. Table S9 lists the selected epochs, the results for every seed and the inference times, and Table S10 breaks the errors down by supervision source.

Ridge predictions on the test set lie between -36.63 and 36.48 minutes, so the ± 60 -minute bound is never active. Combined with the fixed MNL-S choice probabilities, the independent neural predictor reaches a departure MAE of

Table S9

Timing models selected by validation departure MAE, evaluated on the 437 test states.

Model	Seed	Epoch	MAE	Median	P90	Parameters	Train (s)	Infer. (μ s)
Soft KL	42	2	8.461	3.355	25.611	24562	29.9	2.00
Soft KL	2026	117	7.048	5.236	15.801	24562	27.3	2.39
Soft KL	7	31	7.315	5.395	16.061	24562	38.1	2.07
CE + KL	42	2	8.455	3.348	25.600	24562	33.1	1.40
CE + KL	2026	108	6.719	5.452	14.114	24562	38.5	2.23
CE + KL	7	36	7.103	5.110	15.388	24562	34.9	1.83
Signed	42	2	8.464	3.355	25.612	24562	33.8	1.91
Signed	2026	35	7.441	5.399	15.143	24562	33.2	1.78
Signed	7	30	7.239	5.255	15.938	24562	30.9	1.75
Dir. + mag.	42	2	8.459	3.339	25.592	24562	30.2	1.67
Dir. + mag.	2026	108	6.952	4.858	14.918	24562	34.6	1.79
Dir. + mag.	7	30	7.238	5.397	16.072	24562	30.2	1.89
MNL + bounded	42	56	10.237	8.226	19.171	555	2.4	0.20
MNL + bounded	2026	31	10.737	7.321	24.947	555	2.4	0.19
MNL + bounded	7	22	12.592	10.687	23.368	555	2.6	0.19
MNL + neural	42	2	8.423	3.282	25.563	18733	19.7	1.30
MNL + neural	2026	116	7.348	5.248	15.745	18733	19.8	1.91
MNL + neural	7	97	7.046	5.231	15.023	18733	23.7	1.82

All timing errors are in minutes. Every run completes 120 epochs and 6,360 updates; Epoch identifies the minimum validation departure-MAE checkpoint. For the modular Students, parameters include the 444 fixed MNL choice coefficients and the timing predictor. Inference time is the median of 30 batches of the 437 encoded states divided by 437, after five untimed calls and with two CPU threads, excluding encoding and transport preparation. Concurrent workload limits cross-run timing comparisons.

Table S10

Departure errors by test source under validation departure-MAE selection.

Model	Source	States	MAE	Median	P90
Soft KL	single-context	226	6.849 $\pm$ 0.667	4.226	18.489
Soft KL	joint	96	8.923 $\pm$ 2.010	5.378	21.530
Soft KL	mechanism	64	5.921 $\pm$ 0.466	4.643	12.608
Soft KL	accessibility	51	10.614 $\pm$ 0.732	5.379	27.664
CE + KL	single-context	226	6.618 $\pm$ 0.888	4.282	17.240
CE + KL	joint	96	8.729 $\pm$ 2.170	5.319	21.548
CE + KL	mechanism	64	5.911 $\pm$ 0.558	4.546	12.851
CE + KL	accessibility	51	10.453 $\pm$ 0.769	5.354	27.870
Signed	single-context	226	6.964 $\pm$ 0.553	4.300	18.324
Signed	joint	96	9.186 $\pm$ 1.845	5.317	21.257
Signed	mechanism	64	5.843 $\pm$ 0.352	4.636	12.183
Signed	accessibility	51	10.621 $\pm$ 0.595	5.529	28.323
Dir. + mag.	single-context	226	6.765 $\pm$ 0.748	4.199	17.986
Dir. + mag.	joint	96	8.869 $\pm$ 2.055	5.262	21.354
Dir. + mag.	mechanism	64	5.851 $\pm$ 0.385	4.345	11.952
Dir. + mag.	accessibility	51	10.675 $\pm$ 0.528	5.560	28.118
MNL + bounded	single-context	226	10.622 $\pm$ 1.429	8.019	22.556
MNL + bounded	joint	96	12.294 $\pm$ 1.512	10.099	24.505
MNL + bounded	mechanism	64	9.447 $\pm$ 2.162	8.102	17.805
MNL + bounded	accessibility	51	13.805 $\pm$ 1.021	10.171	28.499
MNL + neural	single-context	226	6.807 $\pm$ 0.672	4.429	17.822
MNL + neural	joint	96	8.977 $\pm$ 1.956	5.312	22.015
MNL + neural	mechanism	64	5.806 $\pm$ 0.517	4.144	12.449
MNL + neural	accessibility	51	10.821 $\pm$ 0.819	4.955	27.857

All errors are in minutes. MAE reports the mean and sample SD across three training seeds; Median and P90 are means of the corresponding per-seed error quantiles. Source rows cover all test states and do not affect training or model selection.

7.61 ± 0.72 minutes. It remains 0.180 minutes above the best joint model, CE+KL selected on departure MAE, with a persona-clustered 95% interval of [0.002, 0.418]. Replacing the ridge predictor thus closes most, though not all, of the timing gap to the joint models.

S2.3. Properties of the learning objectives

The signed and direction+magnitude objectives add response penalties to soft KL and place no weight on the hard-label cross-entropy, which only CE+KL includes. The persona, mechanism and accessibility supervision used to adapt SA-Student plays no part in the controlled comparison.

For a freely fitted state with common probability support and equal CE and KL weights,

$$\arg \min_{\mathbf{p}} \{ \text{CE}(a_T, \mathbf{p}) + D_{\text{KL}}(\mathbf{p}_T \| \mathbf{p}) \} = \frac{\mathbf{p}_T + \text{onehot}(a_T)}{2}. \quad (\text{S4})$$

This optimum sharpens the Teacher distribution, and when the Teacher's leading mode is the same in both states of a pair, it halves the idealized probability difference. The fitted networks behave in the opposite way. On the 334 test pairs with an unchanged leading mode, the L_1 magnitudes of the responses, 0.1555, 0.1662 and 0.1876 under soft KL for the three seeds, rise to 0.1810, 0.1910 and 0.2280 under CE+KL. Parameter sharing and finite optimization thus shape the fitted response in ways the single-state solution does not predict.

For Teacher response $t > 0$ and Student response $s \in [-t, 0]$, equally normalized direction and magnitude terms satisfy

$$[-s]_+ + |t - |s|| = t. \quad (\text{S5})$$

The direction term is normalized by the number of available modes with a non-negligible Teacher response, and the magnitude term by the number of all available modes. The loss is flat over this interval whenever the two counts coincide, which holds for 357 of the 398 test pairs when each pair is evaluated alone, although the composition of a batch can change it. In a three-mode example with baseline (0.4, 0.3, 0.3), a Teacher change of (0.2, -0.1, -0.1) and a Student change of (-0.1, 0.05, 0.05), the response gradients cancel while the endpoint losses remain active. A mode with a negligible Teacher response alters the normalization and removes the flat region. Figure S1 illustrates the loss surface and the sensitivity to individual personas.

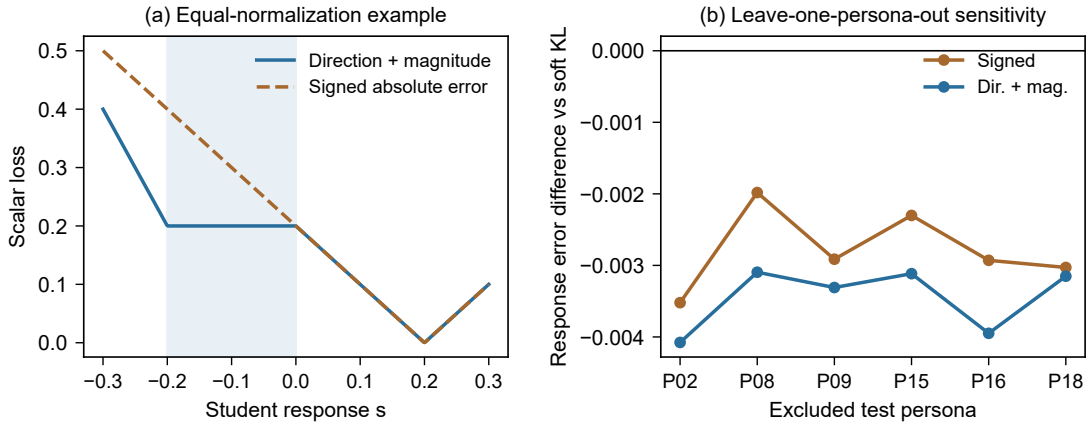


Figure S1: Loss and persona sensitivity. Left: the equal-normalization direction+magnitude penalty and signed absolute error for a Teacher response of 0.2. Right: paired-objective response errors relative to soft KL after omitting each test persona from evaluation. Points average three fitted seeds; models are not retrained.

S2.4. Response robustness

Tables S11–S13 break the response errors down by supervision source, intervention, persona and seed. When each test persona is removed in turn, direction+magnitude remains better than soft KL, with seed-averaged differences

between -0.00408 and -0.00310 and an improvement in all 18 combinations of seed and removed persona. Signed response improves on the seed average after every removal and in 14 of the 18 individual combinations.

Table S11

Where response errors arise in the controlled comparison.

Grouping	Group	Pairs	Soft KL	CE + KL	Signed	Dir. + mag.
source	accessibility	40	0.12287	0.12834	0.11296	0.11947
source	joint	96	0.06726	0.07674	0.06340	0.06211
source	single-context	214	0.06008	0.06523	0.05923	0.05762
source	mechanism	48	0.06987	0.07292	0.06666	0.06540
persona	P000002	70	0.07917	0.08605	0.07992	0.07869
persona	P000008	57	0.06081	0.06934	0.05331	0.05527
persona	P000009	72	0.07308	0.07613	0.07095	0.06901
persona	P000015	58	0.05493	0.06483	0.04941	0.04955
persona	P000016	70	0.07682	0.08440	0.07479	0.07574
persona	P000018	71	0.06689	0.06810	0.06530	0.06209

Values are response errors averaged over the three training seeds. Each grouping independently partitions the same 398 response pairs; its rows must not be added to those of another grouping. Mechanism contrasts retain their constructed label/attribute intervention definition.

Table S12

Controlled response error by intervention definition.

Intervention	Pairs	Soft KL	CE + KL	Signed	Dir. + mag.
Congestion: context only	8	0.05091	0.05784	0.05093	0.04885
Congestion: attributes only	8	0.04187	0.04275	0.04199	0.04152
Congestion: both	8	0.05253	0.06909	0.05570	0.05503
Fare	36	0.04100	0.04289	0.04235	0.04096
Fare + congestion	24	0.03850	0.03984	0.03634	0.03619
Fare + PT delay	24	0.07456	0.07840	0.07016	0.07083
Parking: context only	8	0.08981	0.09016	0.08602	0.08283
Parking: attributes only	8	0.09207	0.09701	0.08363	0.08431
Parking: both	8	0.09204	0.08070	0.08167	0.07987
Parking price	36	0.05709	0.05605	0.05375	0.05385
Congestion	48	0.04134	0.04474	0.04077	0.03972
Congestion + road disruption	24	0.05745	0.07450	0.05266	0.05111
Congestion + rain	24	0.09853	0.11422	0.09445	0.09030
Road disruption	12	0.06587	0.08475	0.06310	0.06346
Supply profile	40	0.12287	0.12834	0.11296	0.11947
PT delay	34	0.06330	0.06899	0.06214	0.06020
Rain	48	0.09163	0.10183	0.09141	0.08755

Each value averages the three validation-KL-selected seeds. Labels follow the definition of each perturbation, not its outcome, and the 17 rows partition all 398 test pairs. Context only, attributes only and both denote mechanism contrasts that change the context description, the level-of-service attributes, or both.

Individual repeated Teacher responses are available for 373 test states, 287 with three queries and 86 with five, which together define 350 complete response pairs. The 64 mechanism states and their 48 pairs lack individual repeats and are excluded from this analysis. The same recomputed targets are used for every Student. Table S14 compares averaging with renormalized medians and with omitting each of the first three queries in turn. Resampling the queries within each state 1,000 times gives intervals for the difference in response error relative to soft KL of $[-0.00319, -0.00195]$ for signed response, $[-0.00366, -0.00270]$ for direction+magnitude and $[-0.01318, -0.00855]$ for MNL-S. These intervals describe sensitivity to the available Teacher queries for fixed fitted models.

Table S13

Per-seed paired response-error differences relative to soft KL, using the validation-KL checkpoints.

Objective	Seed	Difference	95% interval	Pairs	Personas
CE + KL	42	0.00742	[0.00153, 0.01469]	398	6
CE + KL	2026	0.00371	[-0.00207, 0.01002]	398	6
CE + KL	7	0.00679	[0.00069, 0.01278]	398	6
Signed	42	-0.00524	[-0.00815, -0.00228]	398	6
Signed	2026	-0.00334	[-0.00941, 0.00038]	398	6
Signed	7	0.00027	[-0.00252, 0.00304]	398	6
Dir. + mag.	42	-0.00499	[-0.00773, -0.00235]	398	6
Dir. + mag.	2026	-0.00441	[-0.00762, -0.00172]	398	6
Dir. + mag.	7	-0.00094	[-0.00342, 0.00113]	398	6

Negative differences favor the listed objective. Intervals use 10,000 paired persona-cluster resamples, retaining all tasks for each of six test personas. They condition on each fitted seed and are exploratory intervals without adjustment across the methods and seeds.

Table S14

Response-error sensitivity to aggregation of the recoverable Teacher supervision repeats.

Aggregation	Soft KL	CE + KL	Signed	Dir. + mag.	MNL-S
Reference mean	0.06922	0.07560	0.06651	0.06592	0.05727
Median, renormalized	0.07097	0.07798	0.06871	0.06770	0.05850
Omit call 1	0.07801	0.08362	0.07549	0.07487	0.06694
Omit call 2	0.07093	0.07654	0.06865	0.06818	0.06157
Omit call 3	0.06860	0.07630	0.06566	0.06514	0.05626

All rows use the same 373 states and 350 response pairs with recoverable calls. Neural entries average three fixed model seeds. Omit-call conditions remove the indicated query from every state; five-call states retain the remaining four calls. Median probabilities are renormalized to sum to one. No model is retrained.

S3. Human response comparisons

S3.1. Full questionnaire samples

Each contrast compares the change in a respondent's stated choice with the change in model probability across the same tasks. Simple contrasts use weights of -1 and $+1$, and the rain-delay interaction combines four tasks. All explicit choices are retained under the availability rule of Section S1. Table S15 gives further choice diagnostics, and Tables S16–S17 give the complete response results.

Uncertainty is estimated from 10,000 respondent resamples shared across models. Each resampled respondent carries all tasks and model predictions, so within-person pairing is preserved. Neural responses average three fitted seeds, and the spread across seeds is reported separately. Nominal intervals have 95% coverage, and Bonferroni-adjusted intervals within each city have 99% coverage for the five Singapore contrasts and 99.375% for the eight Shanghai contrasts, applied separately for each model. The intervals are conditional on the fitted models. An interval that includes zero reflects uncertainty about the mean bias and does not establish equivalence.

Table S15

Additional questionnaire-task diagnostics.

Model	Balanced accuracy (%)	NLL	Brier
<i>Singapore</i>			
SA-Student	72.8	0.85	0.51
MNL-S	56.3	0.59	0.35
Soft KL	62.7 $\pm$ 2.6	0.65 $\pm$ 0.06	0.38 $\pm$ 0.03
CE+KL	68.6 $\pm$ 2.5	0.75 $\pm$ 0.05	0.43 $\pm$ 0.03
Signed response	62.4 $\pm$ 1.9	0.64 $\pm$ 0.05	0.37 $\pm$ 0.03
Direction+magnitude	60.6 $\pm$ 1.0	0.66 $\pm$ 0.10	0.38 $\pm$ 0.06
<i>Shanghai</i>			
SA-Student	71.7	0.55	0.32
MNL-S	65.9	0.47	0.27
Soft KL	64.3 $\pm$ 5.0	0.54 $\pm$ 0.04	0.32 $\pm$ 0.03
CE+KL	67.3 $\pm$ 3.8	0.57 $\pm$ 0.03	0.34 $\pm$ 0.03
Signed response	64.8 $\pm$ 4.7	0.52 $\pm$ 0.03	0.31 $\pm$ 0.02
Direction+magnitude	64.7 $\pm$ 3.9	0.52 $\pm$ 0.03	0.30 $\pm$ 0.02

Singapore: 332 respondents/3,320 explicit choices; Shanghai: 321/3,208, with two answers of the no-suitable-mode type not scored. Choices of modes the model treats as unavailable are scored, so NLL uses a probability floor of $\epsilon = 10^{-8}$. Accuracy, PT-share bias and mean absolute response bias are given in Table 4 of the main article.

Table S16: Singapore probability responses and stated-choice responses under the trained availability rule.

Model	Contrast	<i>n</i>	Human	Model	Bias	Adjusted interval
SA-Student	PT access (PT)	332	-24.40	-14.42	9.98	[3.01, 16.93]
SA-Student	Rain (bike + walk)	332	-21.99	-19.70	2.29	[-3.40, 8.46]
SA-Student	Fare (PT)	332	-0.60	2.10	2.70	[-4.33, 9.64]
SA-Student	Delay (PT)	332	-12.05	-14.86	-2.81	[-9.81, 4.28]
SA-Student	Roadworks (car)	332	-3.92	-6.09	-2.17	[-7.23, 2.84]
MNL-S	PT access (PT)	332	-24.40	13.85	38.25	[31.03, 45.30]
MNL-S	Rain (bike + walk)	332	-21.99	-9.28	12.71	[7.27, 18.29]
MNL-S	Fare (PT)	332	-0.60	-0.58	0.03	[-6.92, 6.97]
MNL-S	Delay (PT)	332	-12.05	-8.74	3.31	[-3.26, 9.83]
MNL-S	Roadworks (car)	332	-3.92	-4.92	-1.00	[-5.75, 3.84]
Soft KL	PT access (PT)	332	-24.40	-9.97	14.43	[7.79, 21.12]
Soft KL	Rain (bike + walk)	332	-21.99	-1.58	20.40	[14.48, 26.62]
Soft KL	Fare (PT)	332	-0.60	-0.17	0.44	[-6.57, 7.36]
Soft KL	Delay (PT)	332	-12.05	-10.26	1.78	[-4.88, 8.38]
Soft KL	Roadworks (car)	332	-3.92	-4.80	-0.88	[-5.61, 3.95]
CE+KL	PT access (PT)	332	-24.40	-23.37	1.03	[-6.22, 8.11]
CE+KL	Rain (bike + walk)	332	-21.99	-4.95	17.04	[11.15, 23.24]
CE+KL	Fare (PT)	332	-0.60	-0.63	-0.03	[-7.02, 6.88]
CE+KL	Delay (PT)	332	-12.05	-5.42	6.63	[-0.17, 13.45]
CE+KL	Roadworks (car)	332	-3.92	-4.92	-1.01	[-5.75, 3.83]
Signed response	PT access (PT)	332	-24.40	-8.12	16.28	[9.64, 22.91]
Signed response	Rain (bike + walk)	332	-21.99	-3.01	18.98	[13.08, 25.15]
Signed response	Fare (PT)	332	-0.60	-0.24	0.36	[-6.65, 7.30]
Signed response	Delay (PT)	332	-12.05	-11.11	0.94	[-5.71, 7.50]
Signed response	Roadworks (car)	332	-3.92	-4.42	-0.51	[-5.25, 4.34]
Direction+magnitude	PT access (PT)	332	-24.40	-10.06	14.34	[7.67, 20.96]
Direction+magnitude	Rain (bike + walk)	332	-21.99	-3.40	18.59	[12.67, 24.75]

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Singapore response comparison (continued)

Model	Contrast	<i>n</i>	Human	Model	Bias	Adjusted interval
Direction+magnitude	Fare (PT)	332	-0.60	-0.77	-0.16	[-7.18, 6.75]
Direction+magnitude	Delay (PT)	332	-12.05	-13.39	-1.34	[-8.20, 5.50]
Direction+magnitude	Roadworks (car)	332	-3.92	-4.37	-0.46	[-5.20, 4.40]

All changes are percentage points. Model responses for controlled objectives average three fitted seeds. Intervals use shared respondent resamples with Bonferroni adjustment over the city's contrasts. Training-seed variation is separate.

Table S17: Shanghai probability responses and stated-choice responses under the trained availability rule.

Model	Contrast	<i>n</i>	Human	Model	Bias	Adjusted interval
SA-Student	Rain (PT)	321	4.67	2.70	-1.98	[-8.31, 4.21]
SA-Student	Delay (PT)	321	-14.64	-30.57	-15.93	[-22.41, -9.21]
SA-Student	Rain $\times$ delay (PT)	320	9.38	7.99	-1.39	[-9.11, 6.20]
SA-Student	Fare (PT)	321	-5.92	1.13	7.05	[0.15, 13.96]
SA-Student	Parking (car)	321	-9.35	-7.62	1.73	[-2.06, 5.70]
SA-Student	Road penalty (car)	321	-9.35	-12.47	-3.13	[-8.47, 1.68]
SA-Student	Walk vs wait (PT)	320	1.88	-11.75	-13.62	[-19.71, -7.62]
SA-Student	Transfer vs wait (PT)	321	2.49	4.21	1.72	[-5.02, 8.09]
MNL-S	Rain (PT)	321	4.67	5.54	0.87	[-5.21, 7.00]
MNL-S	Delay (PT)	321	-14.64	-8.55	6.09	[0.22, 12.39]
MNL-S	Rain $\times$ delay (PT)	320	9.38	1.45	-7.93	[-15.54, -0.38]
MNL-S	Fare (PT)	321	-5.92	-0.41	5.51	[-1.35, 12.38]
MNL-S	Parking (car)	321	-9.35	-9.45	-0.11	[-3.87, 3.69]
MNL-S	Road penalty (car)	321	-9.35	-9.50	-0.15	[-5.32, 4.65]
MNL-S	Walk vs wait (PT)	320	1.88	-6.70	-8.58	[-14.81, -2.51]
MNL-S	Transfer vs wait (PT)	321	2.49	-1.15	-3.64	[-10.94, 3.14]
Soft KL	Rain (PT)	321	4.67	3.17	-1.50	[-7.91, 4.91]
Soft KL	Delay (PT)	321	-14.64	-18.59	-3.94	[-9.90, 2.30]
Soft KL	Rain $\times$ delay (PT)	320	9.38	0.40	-8.98	[-16.51, -1.38]
Soft KL	Fare (PT)	321	-5.92	0.23	6.15	[-0.77, 13.00]
Soft KL	Parking (car)	321	-9.35	-4.62	4.73	[0.87, 8.91]
Soft KL	Road penalty (car)	321	-9.35	-9.93	-0.58	[-5.74, 4.21]
Soft KL	Walk vs wait (PT)	320	1.88	-7.70	-9.58	[-15.76, -3.52]
Soft KL	Transfer vs wait (PT)	321	2.49	-17.25	-19.74	[-26.48, -13.43]
CE+KL	Rain (PT)	321	4.67	4.49	-0.19	[-6.50, 6.19]
CE+KL	Delay (PT)	321	-14.64	-19.38	-4.74	[-10.73, 1.59]
CE+KL	Rain $\times$ delay (PT)	320	9.38	1.67	-7.71	[-15.30, -0.12]
CE+KL	Fare (PT)	321	-5.92	-0.08	5.83	[-1.08, 12.74]
CE+KL	Parking (car)	321	-9.35	-6.71	2.63	[-1.21, 6.65]
CE+KL	Road penalty (car)	321	-9.35	-11.15	-1.80	[-7.04, 2.97]
CE+KL	Walk vs wait (PT)	320	1.88	-8.95	-10.82	[-16.93, -4.80]
CE+KL	Transfer vs wait (PT)	321	2.49	-31.39	-33.88	[-40.58, -27.36]
Signed response	Rain (PT)	321	4.67	3.44	-1.23	[-7.64, 5.16]
Signed response	Delay (PT)	321	-14.64	-17.15	-2.51	[-8.48, 3.75]
Signed response	Rain $\times$ delay (PT)	320	9.38	1.28	-8.10	[-15.66, -0.56]
Signed response	Fare (PT)	321	-5.92	-0.43	5.49	[-1.42, 12.33]
Signed response	Parking (car)	321	-9.35	-5.92	3.42	[-0.42, 7.49]
Signed response	Road penalty (car)	321	-9.35	-9.75	-0.41	[-5.56, 4.36]
Signed response	Walk vs wait (PT)	320	1.88	-8.88	-10.76	[-16.97, -4.71]
Signed response	Transfer vs wait (PT)	321	2.49	-22.18	-24.68	[-31.38, -18.29]
Direction+magnitude	Rain (PT)	321	4.67	3.39	-1.28	[-7.67, 5.09]
Direction+magnitude	Delay (PT)	321	-14.64	-16.98	-2.34	[-8.28, 3.87]
Direction+magnitude	Rain $\times$ delay (PT)	320	9.38	0.85	-8.52	[-16.07, -0.96]
Direction+magnitude	Fare (PT)	321	-5.92	-0.70	5.22	[-1.68, 12.09]

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Shanghai response comparison (continued)

Model	Contrast	<i>n</i>	Human	Model	Bias	Adjusted interval
Direction+magnitude	Parking (car)	321	-9.35	-5.16	4.19	[0.34, 8.33]
Direction+magnitude	Road penalty (car)	321	-9.35	-9.12	0.22	[-4.91, 4.98]
Direction+magnitude	Walk vs wait (PT)	320	1.88	-7.88	-9.75	[-15.93, -3.71]
Direction+magnitude	Transfer vs wait (PT)	321	2.49	-16.63	-19.12	[-25.93, -12.77]

All changes are percentage points. Model responses for controlled objectives average three fitted seeds. Intervals use shared respondent resamples with Bonferroni adjustment over the city's contrasts. Training-seed variation is separate. Table S18 reports switches in both directions within each Singapore task pair. The exact McNemar tests, which are not adjusted for multiplicity, test whether switching into and out of the target mode group is equally frequent. They concern the human data alone and are separate from the comparisons with models.

Table S18

Singapore within-person stated-choice effects ($N = 332$). Counts refer to the primary mode group in each task pair. Changes are percentage points.

Pair (tasks)	Outcome	Baseline	Intervention	Lost	Gained	Δ pp	Exact <i>p</i>
A (1/6)	PT	218	137	87	6	-24.4	< 0.001
W (3/7)	Active	83	10	75	2	-22.0	< 0.001
D (5/9)	PT	239	199	61	21	-12.0	< 0.001
F (8/2)	PT	222	220	43	41	-0.6	0.913
R (10/4)	Car	35	22	27	14	-3.9	0.060

A: access and transfers; W: rain; D: PT delay; F: fare; R: road works. Active combines bicycle and walking. Lost/gained count respondents leaving/entering the primary mode group. Two-sided exact McNemar tests condition on within-person discordances; five tests are unadjusted.

Baseline/intervention percentages are 100 times the counts divided by 332.

S3.2. Contemporary Teacher on matched tasks

The Teacher comparison uses 24 respondents per city, drawn in turn from strata defined by age and car ownership (random seed 20260921). Each respondent contributes ten tasks, and each task receives three valid Teacher queries, giving 240 explicit choices per city. Every contrast has 24 complete respondents, and the subsets contain no answers of the no-suitable-mode type and no choices of modes that the model treats as unavailable.

The Teacher prompts contain the same numerical and categorical attributes and the same ownership and access rule as the Student inputs, but not the recorded human choices. Persona identifiers are replaced by neutral codes, task and context identifiers by constants, and origin and destination types by an unspecified value. The Students do not use any of these metadata. Monetary values are given without currency names. The prompt is the one used to label the accessibility supervision of SA-Student, which explains PT feasibility and the components of door-to-door time. The reference values and the private-car representation follow Section S1.

All successful requests returned the model field `deepseek-v4-pro` from the official DeepSeek service. Requests specified a temperature of 0.2 and a budget of 16,384 tokens, without changing the default reasoning settings. According to the provider's documentation, the default reasoning mode uses high effort and ignores the temperature parameter,¹ so the temperature setting may not have controlled sampling. These queries were made after the Students had been trained and are separate from the supervision used to train them.

The three probability vectors for each state are averaged. Table S19 gives choice metrics for the subsets, and Table S20 gives the human and Teacher responses together with the variation across repeated queries. Neural accuracy is computed for each fitted seed before averaging.

For a contrast with mean responses H , T and S for people, Teacher and Student, the signed differences satisfy $S - H = (T - H) + (S - T)$. The intervals use 10,000 respondent resamples (seed 20260922) with identical draws for all models and terms. Table 5 in the main article decomposes the discrepancy of SA-Student, and Table S21 gives the Student-Teacher differences for the other three Students. Differences between Teacher and people appear in both of these tables and in Table S20. Nominal intervals, Student-human differences and changes in absolute bias are provided in the supplementary data files (Section S7). Across the 240 states in each city, the mean within-state SD of PT

¹DeepSeek API documentation, https://api-docs.deepseek.com/guides/thinking_mode/, accessed 22 September 2026.

Table S19

Choice metrics on the Teacher comparison subsets. Each city contributes 24 people and 240 explicit choices. Neural families show the arithmetic mean and sample SD of three separately scored training seeds.

Model	Accuracy (%)	Brier score	PT-share bias (pp)
<i>Singapore</i>			
Current Teacher	75.42	0.356	-6.21
SA-Student	61.67	0.549	-29.70
MNL-S	65.00	0.428	0.51
Soft KL	65.42 $\pm$ 2.32	0.442 $\pm$ 0.049	-17.98 $\pm$ 2.55
CE+KL	63.75 $\pm$ 0.72	0.488 $\pm$ 0.049	-23.26 $\pm$ 2.75
Signed response	65.28 $\pm$ 2.68	0.437 $\pm$ 0.053	-16.32 $\pm$ 3.75
Direction+magnitude	64.86 $\pm$ 4.17	0.443 $\pm$ 0.065	-16.68 $\pm$ 5.70
<i>Shanghai</i>			
Current Teacher	85.00	0.233	4.32
SA-Student	76.67	0.342	-18.71
MNL-S	78.33	0.294	-2.53
Soft KL	76.39 $\pm$ 0.24	0.329 $\pm$ 0.027	-12.54 $\pm$ 7.59
CE+KL	74.44 $\pm$ 2.51	0.350 $\pm$ 0.015	-14.90 $\pm$ 6.75
Signed response	77.92 $\pm$ 2.73	0.319 $\pm$ 0.023	-11.20 $\pm$ 6.03
Direction+magnitude	77.92 $\pm$ 1.10	0.317 $\pm$ 0.022	-10.74 $\pm$ 5.80

Teacher choice metrics use the mean probabilities from three successful queries per state. SA-Student and MNL-S are single fitted models; their rows have no training-seed SD. For the four neural objectives, probabilities are not averaged across seeds before computing accuracy or Brier score. PT-share bias is predicted probability minus the human PT indicator. Neither subset contains an answer of the no-suitable-mode type or a choice of a mode the model treats as unavailable.

Table S20

Human and contemporary Teacher responses on the 24-person subsets. The SD describes variation across three repeated queries, not respondent uncertainty.

Contrast	Human response (pp)	Teacher response (pp)	Query-repeat SD (pp)
<i>Singapore</i>			
Poorer PT access	-37.50	-15.40	1.13
Rain (active modes)	-29.17	-12.49	0.43
PT fare	+4.17	-5.40	1.82
PT delay	-12.50	-10.56	0.61
Road penalty	-20.83	-8.35	0.91
<i>Shanghai</i>			
Rain	+20.83	+0.64	0.46
PT delay	-8.33	-8.17	0.72
Rain $\times$ delay	-4.17	+1.14	0.33
PT fare	+0.00	+0.03	0.23
Parking	-8.33	-1.31	0.78
Road penalty	-12.50	-8.72	0.88
Walk versus wait	-16.67	-0.36	0.86
Transfer versus wait	-16.67	+1.26	0.75

Each repeated-query response combines the k -th query of every task and respondent, and queries are independent across states. The SD is computed over the three resulting responses. The respondent-bootstrap intervals in the decomposition tables describe a different source of variation.

Table S21

Student–Teacher response differences on the matched subsets.

Contrast	MNL-S – Teacher	Soft KL – Teacher	Direction+magnitude – Teacher
<i>Singapore (n = 24)</i>			
Poorer PT access	+34.8 [24.7, 44.8]	+5.5 [0.2, 11.3]	+5.5 [0.6, 11.0]
Rain	+3.9 [−0.2, 8.8]	+10.5 [4.6, 17.3]	+9.4 [3.7, 16.1]
Fare	+4.7 [1.0, 9.3]	+5.8 [1.6, 10.7]	+5.3 [1.1, 10.2]
PT delay	+0.5 [−3.0, 4.1]	+0.3 [−4.6, 5.8]	−2.0 [−9.1, 5.6]
Road penalty	+1.2 [−2.0, 5.2]	+1.5 [−2.0, 5.4]	+2.0 [−1.3, 6.0]
<i>Shanghai (n = 24)</i>			
Rain	+5.1 [2.8, 7.6]	+2.4 [−3.6, 6.7]	+2.8 [−3.0, 6.8]
PT delay	−0.7 [−4.1, 2.9]	−10.9 [−15.3, −5.6]	−9.0 [−13.0, −4.3]
Rain × delay	+0.6 [−2.9, 4.2]	−0.7 [−4.5, 3.7]	−0.1 [−3.8, 4.3]
Fare	−0.5 [−1.7, 0.5]	+0.2 [−1.2, 1.4]	−0.7 [−2.0, 0.5]
Parking	−6.8 [−14.4, −0.9]	−2.2 [−5.2, 0.2]	−2.8 [−6.4, 0.0]
Road penalty	+0.5 [−4.1, 4.8]	−0.3 [−4.2, 2.3]	+0.6 [−3.0, 3.4]
Walk versus wait	−6.3 [−9.0, −3.9]	−7.8 [−9.6, −5.8]	−7.8 [−9.4, −6.0]
Transfer versus wait	−3.9 [−13.4, 4.3]	−19.3 [−21.3, −17.1]	−18.4 [−20.4, −16.2]

Entries are percentage points with within-city family-adjusted respondent-bootstrap intervals (five Singapore or eight Shanghai contrasts), conditional on the fitted Students and three-call Teacher means. Teacher–human differences are reported in Supplementary Table S20 and Table 5 of the main article. Nominal intervals, Student–human differences and changes in absolute bias are given in the supplementary data files.

probability is 4.47 percentage points in Singapore and 2.95 in Shanghai, and the mean SD of departure adjustment is 2.05 and 1.47 minutes. Repeated queries are independent across states, so these values describe variability in the Teacher’s answers and are distinct from respondent uncertainty.

S3.3. Teacher prompt and metadata sensitivity

A small diagnostic uses the first two selected respondents in each city, with all ten tasks and three queries per condition, and keeps the ownership and access rule of the Students. One comparison replaces a generic prompt with the accessibility-aware prompt while metadata are held neutral. The other replaces identifiers that reveal the city with neutral values while the generic prompt is held fixed, without removing all metadata fields.

Across these 40 states, the change of prompt shifts mean PT probability by +1.04 points and the neutral metadata shift it by −1.06 points (Table S22). Both prompt conditions used the same token budget, whereas some requests in the metadata comparison were repeated with different budgets. Table S23 reports the variation across repeated queries.

Table S22

Descriptive matched diagnostics using the first two preselected people per city (four people, 40 states, three queries per state in each condition). Changes are right minus left under the comparisons defined in the text.

Comparison	Scope	Mean state probability L_1	PT change (pp)
System prompt	Singapore	0.0767	+1.55
System prompt	Shanghai	0.0327	+0.53
System prompt	Pooled four people	0.0547	+1.04
Unused metadata	Singapore	0.0843	−1.00
Unused metadata	Shanghai	0.0343	−1.12
Unused metadata	Pooled four people	0.0593	−1.06

All task groups and all three query repeats are complete; no person was replaced and no human answer was used in this diagnostic. Probability vectors are averaged over repeats before calculating the state-level differences. The pooled result weights the two cities equally, with 20 states each. The differences are descriptive and carry no confidence intervals.

Table S23

Within-state Teacher query variability in the matched four-person diagnostic. Entries are the mean sample SD of PT probability across three calls, in percentage points.

Condition	Singapore	Shanghai	Pooled
Generic / city metadata / original support	4.79	1.65	3.22
Generic / neutral metadata / original support	3.99	2.36	3.18
Accessibility / neutral / original support	3.94	1.84	2.89

Variability is computed on successful answers only and describes the repeatability of the Teacher under the recorded request settings.

S3.4. Shanghai supply-input sensitivity

In the network-based condition, each respondent is assigned the origin–destination pair closest to a 6-km target among 150 sampled origins, which yields 61 distinct pairs for the 321 respondents. Twenty further assignments repeat this rule with independent random draws. Twenty permutations exchange complete baseline supply profiles, together with their origin–destination pair and departure time, between respondents while personal characteristics stay with each respondent. All ten tasks apply their specified changes to the assigned baseline. Human answers play no part in the assignment.

Table S24 scores all 3,208 explicit choices under the model’s availability rule. Relative to the reference inputs, network-based inputs raise accuracy by 2.804 percentage points (respondent-paired 95% interval [2.055, 3.583]) and change the Brier score by -0.0452 ($[-0.0547, -0.0351]$). Gains are averaged over tasks within each respondent and then over respondents. The mean gain is 2.564 points under fresh assignments and 2.519 points after exchanging supply profiles. That the gain survives the exchange suggests that it stems largely from a general shift in the numerical inputs. The assigned trips are constructed scenarios, not observations of respondents’ journeys.

Table S24

Shanghai assigned-input sensitivity under the ownership-conditioned model support.

Input encoding	Runs	Accuracy (%)	Paired gain (pp)	Brier
			mean $\pm$ SD [range]	
Fixed reference	1	80.46	–	0.3241
Assigned OD	1	83.26	2.804	0.2789
Fresh OD assignment	20	83.02 ± 0.21	2.564 ± 0.214	0.2780 ± 0.0021
			[2.181, 2.866]	
Whole-vector permutation	20	82.98 ± 0.16	2.519 ± 0.163	0.2787 ± 0.0020
			[2.212, 2.804]	

Every run retains 321 respondents and 3,208 explicit labels, including two structurally unsupported bicycle labels. Accuracy and Brier use all explicit choices; gains average paired changes within each respondent. Dispersion and ranges describe assignments, not respondent uncertainty.

S4. Supply-adapted Student

SA-Student adapts a previously trained joint neural Student to six additional PT-supply fields. The existing weights, categorical vocabularies and normalization are kept. The input weights for the new fields start at zero, and their normalization is fitted on the accessibility training states. The adapted model has 24,562 parameters. The accessibility supervision contains 336 states, divided into 234 for training, 51 for validation and 51 for testing, with disjoint pools of 79, 14 and 20 origin–destination pairs and of 28, six and six personas. Its targets average three Teacher queries for 89 states and five for 247, 1,502 valid queries in total.

SA-Student applies the direction and magnitude terms of the main text to the modes available in the perturbed state, whereas the controlled comparison uses the modes available in either state. Mechanism supervision matches the responses from a baseline to variants that change the context together with the attributes, the context alone or the attributes alone. Persona supervision matches differences between travelers making the same trip, and accessibility supervision matches the PT response between the best and worst accessibility classes for a fixed traveler and trip.

Each epoch draws 104 single-context pairs and 52 each of joint-context pairs, persona pairs, accessibility states and accessibility-curve pairs, limited by the size of the available pools, while mechanism updates use the available groups

Table S25

Specification and selection of the supply-adapted Student.

Item	SA-Student configuration
Initialization	Previously trained joint Student; weights of the six added supply fields set to zero
Parameters	24,562
Accessibility split	234 / 51 / 51 states (training / validation / test)
State-level objective	CE + KL for mode choice; Huber loss with one-minute transition for departure
Additional supervision	Direction/magnitude, mechanism, persona and accessibility contrasts
Supervision sources	Single-context : joint context : mechanism : accessibility = 2:1:1:1, with additional persona and accessibility-curve updates
Optimizer	Adam; learning rate 1.25×10^{-4} ; weight decay 10^{-4}
Batch sizes	32 states/pairs; eight mechanism quadruplets
Training schedule	Seed 42; maximum 120 epochs; patience 15
Selection	Minimum KL on accessibility validation states, subject to at most a one-point loss in single-context accuracy and at most 10% increases in single-context KL and in KL on joint contexts seen in training, relative to initialization
Selected epoch	17
Evaluation	Weights fixed in the stated-choice and simulation experiments

of four states. All active loss terms have unit weight, and each type of update keeps its own reduction by sum or mean. The supplementary data files describe these settings and include the accessibility prompt. Because its initialization, additional supervision and adaptation differ from the controlled objectives of Section S2, SA-Student is not treated as part of the matched comparison of objectives. The predecessor checkpoint and its training history are retained in the corresponding model release.

S5. Simulation experiments

S5.1. Helsinki response through assignment and routing

The Helsinki experiment uses a fixed population of 1,000 travelers with fixed origin–destination pairs, HSL timetables and an OSM road network. The perceived-delay condition changes the input to the behavioral model while the physical timetable stays the same. Table S26 summarizes the fitted models, assignment rules and paired comparisons. The soft-KL and direction+magnitude Students are the three models per objective selected by validation KL, and MNL-S is combined with each of its three independently trained neural timing predictors.

PT use is traced through predicted probabilities, feasibility-adjusted probabilities, assigned modes, routed modes and simulated boarding, with records matched by person. A traveler counts as a simulated PT user after boarding a PT vehicle before the outbound destination, whether or not the journey is later completed. All responses use the same 1,000 people as denominator, whereas the share of assigned PT users who board is computed over the number assigned to PT. Completion, boardings and legs are reported as separate outcomes. Table S27 gives every intermediate stage, and Table S28 contrasts signed averages with absolute gaps. Trips are routed at the departure time adjusted by each model. For SA-Student, routing at the adjusted rather than the desired departure time changes the itinerary of 279 travelers at baseline and 313 under delay, while the same 684 people have a feasible PT route at both departure times in each condition. Itineraries can thus change without any change in feasibility. The supplementary data files list all 70 baseline–delay comparisons with their paired-person intervals, at the precision reported in the tables. Section S7 describes which details of the routing procedure these summaries document.

Table S26

Design of the Helsinki simulation experiment.

Quantity	Configuration
Initial population	1,000 fixed people and OD pairs
Conditions	Baseline and 15-minute perceived-delay input; physical timetable fixed
Fitted models	10: SA-Student; three each of soft KL, direction+magnitude and MNL-S with timing
Assignment rules	Deterministic, stochastic, and feasibility-constrained stochastic assignment
Assignment seeds	42, 2026 and 7, with common draws across paired conditions
Runs	14 per fitted model; 140 runs, 70 paired baseline–delay comparisons
Simulation	Flow and storage capacity factors of 0.3; one MATSim iteration

Walking and cycling trips use speeds derived from the road links. Walking legs outside the network, such as access to stops, use 1.39 m/s.

Table S27

Helsinki PT responses at every stage from prediction to simulation, averaged over assignment repetitions within each fitted model and then over models.

Model	Assignment	Predicted	Adjusted	Assigned	Routed	Simulated
Direction+magnitude	Constrained	-6.77	-6.84	-7.22	-7.22	-7.22
Direction+magnitude	Deterministic	-6.77	-6.77	-8.10	-8.63	-8.63
Direction+magnitude	Stochastic	-6.77	-6.77	-7.01	-7.12	-7.12
MNL-S + neural time	Constrained	-6.84	-6.10	-6.67	-6.67	-6.67
MNL-S + neural time	Deterministic	-6.84	-6.84	-7.30	-6.90	-6.90
MNL-S + neural time	Stochastic	-6.84	-6.84	-7.40	-6.63	-6.63
SA-Student	Constrained	-13.10	-12.83	-14.17	-14.17	-14.17
SA-Student	Deterministic	-13.10	-13.10	-13.00	-12.30	-12.30
SA-Student	Stochastic	-13.10	-13.10	-14.60	-14.17	-14.17
Soft KL	Constrained	-6.97	-6.60	-7.06	-7.06	-7.06
Soft KL	Deterministic	-6.97	-6.97	-7.60	-8.03	-8.03
Soft KL	Stochastic	-6.97	-6.97	-7.34	-6.97	-6.97

All entries are percentage points with the same 1,000 initial people per run. The main table separates training variation from sampling variation.

Results for individual runs are given in the supplementary data files.

Table S28

Cancellation of signed gaps between predicted and simulated responses in Helsinki.

Model	Policy	Models	Pairs	$ \bar{g} $	$ \bar{g}_m $	$ \bar{g}_{ms} $
SA-Student	Deterministic	1	1	0.795	0.795	0.795
SA-Student	Stochastic	1	3	1.072	1.072	1.402
SA-Student	Constrained	1	3	1.072	1.072	1.268
Soft KL	Deterministic	3	3	1.061	1.179	1.179
Soft KL	Stochastic	3	9	0.005	1.453	1.508
Soft KL	Constrained	3	9	0.084	1.380	1.419
Direction+magnitude	Deterministic	3	3	1.865	1.865	1.865
Direction+magnitude	Stochastic	3	9	0.354	0.906	0.961
Direction+magnitude	Constrained	3	9	0.454	0.872	0.883
MNL-S + neural time	Deterministic	3	3	0.061	0.061	0.061
MNL-S + neural time	Stochastic	3	9	0.206	0.206	0.780
MNL-S + neural time	Constrained	3	9	0.173	0.173	0.813

All gaps are simulated minus predicted response, in percentage points. The first measure takes the absolute value after averaging signed gaps across sampling and training seeds. The second first averages sampling within each fitted model, then averages the absolute model gaps. The third averages absolute gaps over all baseline–delay run pairs. Opposite signed gaps can cancel only in the first measure across models; sampling effects can additionally cancel within the second. All three are summaries of the same runs.

S5.2. Active-mode speed sensitivity in Shanghai

The Shanghai experiment holds the 321 people, their assigned modes, departure times and plans fixed, and compares active-mode speeds derived from the network links with caps of 1.34 m/s for walking and 4.17 m/s for cycling. The timetable is the same in the baseline and perceived-delay conditions. PT users number 224 at baseline and 134 under delay, and every person completes both journeys.

With network-derived speeds, mean outbound times are 19.68 minutes at baseline and 15.23 minutes under perceived delay. With speed caps, the corresponding means are 33.59 and 47.63 minutes. The paired difference is therefore -4.44 minutes ($[-5.06, -3.84]$) under link speeds and $+14.04$ minutes ($[11.18, 17.05]$) under speed caps (Figure S2). Delay moves 90 travelers away from PT, and whether this shortens or lengthens the average journey depends on the speeds assumed for walking and cycling. The times are model outputs and have not been compared with observed trip durations.

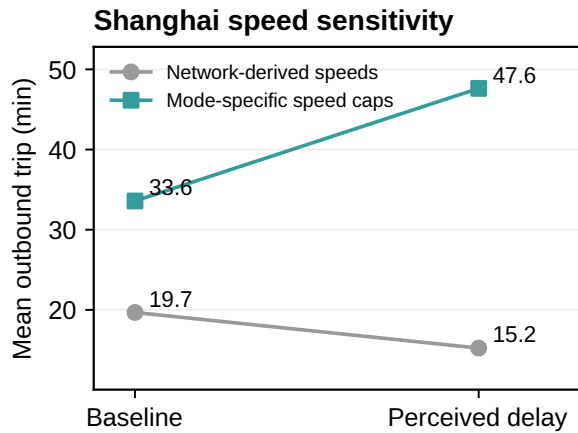


Figure S2: Sensitivity of Shanghai journey times to the speeds of walking and cycling in the simulation. People, assigned modes, departure times and physical timetables are identical across speed settings. Values are mean durations of completed outbound trips.

S5.3. Singapore population responses and journey measures

The Singapore scenario experiment simulates 10,000 synthetic travelers on unchanged physical supply. Scenario C0 is the baseline, C1 introduces heavy rain, C2 a 50% fare increase and C3 a perceived PT delay of 15 minutes. C4 represents a road disruption that adds 20 minutes of car travel time and 20 minutes of reliability delay, and C5 combines rain with PT delay. Populations drawn with seeds 2026, 42 and 7 vary the composition of travelers, and MATSim uses seed 4711. Table S29 gives the exact shares shown in Figure 1 of the main article, and Table S30 their sensitivity to population composition. Journey measures for each person run from the first departure to the first activity other

Table S29

Singapore absolute Student decision shares (percent) and MATSim consequences. Time is the completed-leg mean, not the mean journey duration per person. Seed 2026, 10,000 agents.

Scenario	Car	PT	Bike	Walk	Boardings	Car VKT (km)	Completed-leg mean (min)
C0	28.3	25.3	34.0	12.4	5613	33,939.0	43.67
C1	37.6	45.3	13.3	3.8	9761	42,777.5	50.86
C2	29.9	25.7	32.6	11.8	5674	35,797.8	43.86
C3	29.0	10.3	35.8	25.0	2474	34,684.6	32.23
C4	1.8	37.5	41.8	19.0	8537	2,943.4	52.71
C5	37.7	18.6	24.5	19.2	4301	42,822.8	38.27

Table S30

Paired population-seed responses from the existing multi-seed summary.

Scenario	Seed	Δ PT pp	Δ car pp
C1	2026	+20.0	+9.3
C1	42	+19.7	+9.2
C1	7	+20.1	+9.2
C2	2026	+0.4	+1.6
C2	42	+0.5	+1.8
C2	7	+0.5	+1.5
C3	2026	-15.1	+0.8
C3	42	-15.3	+0.6
C3	7	-15.7	+0.6
C4	2026	+12.1	-26.5
C4	42	+11.7	-26.1
C4	7	+12.3	-26.5
C5	2026	-6.7	+9.4
C5	42	-6.4	+9.3
C5	7	-6.9	+9.1

than home or an interaction point created by routing, such as a transfer. Table S31 reports people, boardings and legs separately. At baseline, 9,741 people complete the outbound journey and 9,062 return home. A completed outbound journey takes 66.43 minutes on average, whereas a completed leg takes 43.67 minutes, because a PT journey can consist of several legs.

Table S32 also reports, for every departing person, the time elapsed until completion or until the end of the 30-hour simulation, a descriptive measure that includes unfinished journeys. Paired differences between scenarios use people who complete both trips and 4,000 paired-person resamples. All times reflect the network-based speeds and the fixed supply.

Table S31

Person and leg counts across six Singapore scenarios.

Scenario	Outbound complete	Return complete	Outbound PT people	Boardings	Unfinished legs	Stuck people	Completed legs
C0	9,741	9,062	2,456	5,613	938	211	29,476
C1	9,546	8,367	4,442	9,761	1,633	366	36,477
C2	9,730	9,044	2,492	5,674	956	221	29,567
C3	9,884	9,600	961	2,474	400	102	24,219
C4	9,647	8,619	3,671	8,537	1,381	285	34,499
C5	9,789	9,268	1,779	4,301	732	183	27,266

Each scenario starts with and departs all 10,000 people. Completion and PT use are person counts. Boardings include transfers and both travel directions; unfinished legs and completed legs are different event-level counts. A stuck person may also have an unfinished leg.

Table S32

Singapore timing with explicit units and completion conditions. All times are in minutes.

Scenario	Completed-leg mean	Outbound mean	Outbound median	Completion-or-horizon mean	Paired people	Paired outbound change versus C0 [95% CI]
C0	43.67	66.43	17.45	88.80	–	–
C1	50.86	98.50	20.19	135.60	9,511	+32.00 [+29.17, +34.69]
C2	43.86	66.88	16.97	90.01	9,708	+0.60 [-0.45, +1.69]
C3	32.23	38.92	15.67	49.78	9,724	-27.68 [-29.86, -25.56]
C4	52.71	94.78	27.08	124.21	9,578	+28.39 [+25.98, +30.83]
C5	38.27	53.33	14.35	71.97	9,695	-13.87 [-15.83, -11.96]

Outbound means and medians condition on the scenario-specific completers in Table S31; the horizon mean retains all 10,000 departed people until completion or the administrative end at simulation hour 30. Paired changes retain only people completing both outbound journeys and use 4,000 paired-person bootstrap resamples (seed 917). The link-speed implementation limits interpretation as calibrated travel time.

S5.4. Feasibility correction and response geometry

For a normalized distribution $\mathbf{p}$ and feasible set F with mass $Z > 0$, conditional renormalization adds mass $1 - Z$ within F and removes mass $1 - Z$ outside it. Hence $\|\mathbf{q} - \mathbf{p}\|_1 = 2(1 - Z)$. Every distribution supported on F must remove all outside mass, and its inside changes have a sum of $1 - Z$; the triangle inequality therefore gives the same lower bound. Conditional renormalization attains it, though the L_1 minimizer need not be unique.

An endpoint-optimal correction need not preserve a paired response. For a fixed feasible set $F = \{1, 2\}$, let baseline probabilities be $(0.4, 0.3, 0.3)$ and perturbed probabilities $(0.4, 0.2, 0.4)$. Their corrections are $(4/7, 3/7, 0)$ and $(2/3, 1/3, 0)$. The probability of mode 1 is unchanged before correction but increases by $2/21$ afterward. This constructed example proves the distinction without making a claim about its empirical prevalence. The main-text decomposition reports the specified correction separately from subsequent execution; it does not attribute every correction-induced response difference to an unavoidable lower bound.

S6. Computational measurements

Inference time for choice and timing together is measured on the 437 test states after feature encoding, using two CPU threads. After five untimed calls, 30 batches are timed, and the median batch time is divided by 437 for each seed (Table S9). MNL-S with neural timing requires 1.68 ± 0.33 microseconds per state, compared with 0.19 ± 0.002 with bounded-linear timing and 1.78 – 2.16 microseconds for the joint neural Students. Feature encoding and route preparation are not included.

The Teacher and SA-Student are compared on the same first 100 states, queried one at a time. The Teacher answers 87 of the 100 requests successfully, taking 86.539 seconds per successful request on average (95% bootstrap interval 76.745–96.550), whereas SA-Student takes 0.272 milliseconds per state on a local processor. Neither figure includes the construction of a full population. Table S33 distinguishes measured values from values scaled to larger populations.

Table S33

Time required for behavioral prediction alone. Teacher latency refers to successful requests, and Student times exclude supply preparation. Values for 10,000 agents are scaled or projected, not measured constructions.

Model	Measurement	Mean latency	10,000 equivalent
Teacher (remote)	100 attempts, 87 successes	86.539 s	240.4 h (projected)
SA-Student sequential	Same first 100 states, one CPU thread	0.272 ms	–
SA-Student sequential	100,000 states, one CPU thread	0.332 ms	3.3 s (scaled)
SA-Student batch 256	20,000 states, 24 CPU threads	25,399 states/s	0.4 s (scaled)

The nested Singapore populations contain 1,000, 10,000, 20,000 and 50,000 people, simulated with flow and storage capacity factors of 0.3 throughout (Table S34). Construction time covers feature preparation, behavioral prediction and plan generation. The value for 1,000 people is the median of three constructions with identical behavioral predictions. Because larger populations also load the network more heavily at unchanged capacity factors, their simulation times do not correspond to calibrated city populations.

Table S34

Population-scale construction and transport outcomes under unchanged Singapore supply, C0, seed 2026. Leg time averages completed legs, and unfinished legs count departures without a matching arrival. Smaller populations are subsets of larger ones, and the 1,000-agent row is the median of three constructions.

Agents	Build (min)	Decisions (s)	MATSim (min)	Boardings	Unfinished legs	Completed-leg mean (min)
1,000	5.9	1.0	2.4	560	98	43.82
10,000	51.1	10.3	2.1	5,613	938	43.67
20,000	98.9	18.8	2.4	11,357	1,956	43.90
50,000	231.8	48.8	3.4	28,440	5,586	48.92

Construction with and without reuse is compared for the same population and scenario. With reuse, route and accessibility calculations that depend only on the network and timetable are taken from an earlier construction, while Student predictions are recomputed. Construction time falls from 3,223.6 to 469.5 seconds in Singapore and from 10,491.8 to 103.2 seconds in Helsinki (Table S35). All 10,000 mode and origin–destination assignments are unchanged, and only two Singapore departures and one Helsinki departure differ, by 0.01 minutes after rounding. All constructions

Table S35

Scenario-construction time for 10,000 agents without (cold) and with (warm) reuse of unchanged network and timetable calculations. Student predictions are recomputed in both cases.

City	Original (s)	Cold (s)	Warm (s)	Cold/warm
Singapore	3,063.1	3,223.6	469.5	6.9
Helsinki	9,507.0	10,491.8	103.2	101.7

ran on a machine with 24 logical CPU cores and 31.4 GB of memory, using PyTorch without GPU acceleration. Each city was measured once with reuse. The 50,000-person construction partly overlapped a preliminary Helsinki run, and one Helsinki delay construction shared the machine with other work, so these timings describe this machine under its recorded workload. The full cost of distillation would also include acquisition and retry charges that were not recorded in full. The retrospective accounting below uses available usage records and does not fill missing charges with zeros. The largest simulated population has 50,000 people, and the separate measurement on 100,000 states concerns inference alone.

S6.1. Retrospective token accounting

The accounting script reads retained repeat and API-call records, deduplicates them by completion ID and exports only aggregate counts and file hashes. At peak rates, a completion with T_h cached input tokens, T_m uncached input tokens and T_o output tokens is repriced as

$$C_{\text{API}} = (0.044T_h + 1.32T_m + 3.96T_o)/10^6 \quad \text{USD.} \quad (\text{S6})$$

Off-peak rates are half these values. These are the provider's rates checked on 25 September 2026, not a currency conversion or recovered invoice. The separately published CNY rates give CNY 700.30–1,400.61 for the same retained metered total. Promotions, account credits, tax treatment and unmetered requests are not reconstructed. The reproducible aggregate ledger is included with the manuscript under `reproduction/cost_audit/`; it contains no participant-level payloads.

The final same-task campaign made 2,880 recorded attempts: 1,440 valid responses and 1,440 HTTP 402 failures without usage. Its metered responses contain 1,409,152 cached and 915,218 uncached input tokens and 3,254,670 output tokens. Unknown charges for failures are not asserted to be zero. The earlier generic-prompt campaign had 1,475 usage-bearing responses, of which 35 were rejected by parsing; their tokens remain in the cost sum. The final campaign consumed an average of 1,614 input and 2,260 output tokens per valid response, which can inform a workload-matched budget but is not a guaranteed token ceiling for other prompts.

The 12 retained controlled-neural training status files report 29.58–32.13 seconds per fit on the recorded CPU setup, or 371.09 seconds summed over the fits. This sum is historical fit time, not elapsed campaign time or the

complete training and adaptation history. No hardware rental or electricity charge was recorded. For reuse, the original acquisition cost is sunk; retraining from frozen targets adds no Teacher calls, while collecting new targets must be budgeted separately.

S7. Model roles, input sources and research materials

S7.1. Model roles and input sources

Table S36 summarizes how the three model families enter the study, and Table S37 the sources of information in the stated-choice inputs. Fitting settings and the numerical task specifications are given in Sections S2, S4 and S1.

Table S36

Model families, their estimation and their role in the study.

Model family	Behavioral specification	Estimation and role
Joint neural Students	Shared representations with separate choice and departure outputs	Four objectives trained under matched conditions isolate the effect of the loss. Probability responses and timing are evaluated with separately selected fits.
MNL-S with separate timing	Linear choice utilities and an independently estimated timing predictor	Same inputs and training states, with a separately regularized choice model. Compares functional specifications and the separation of outcomes.
Supply-adapted Student	Adapted joint neural model with additional PT attributes	Differs in initialization, objective, auxiliary supervision and treatment of availability. Evaluated as an additional model, not as part of the matched loss comparison.

Table S37

Sources of information in the stated-choice model inputs.

Source	Information used	Interpretation
Shown in the questionnaire	Singapore time and cost tables; the Shanghai PT fare of 6 yuan, parking fee of 30 yuan and one transfer	Attributes presented to respondents, with different numerical detail in the two surveys.
Reported by respondents	Personal characteristics supplied by respondents and covered by the model's input mapping	Observed inputs, although not every model attribute was collected.
Specified by the researchers	Characteristics not collected, and Shanghai durations and time components not shown in the tasks	Analysis assumptions. The twelve Singapore and eight Shanghai profiles examine selected alternatives rather than a full uncertainty distribution.

S7.2. Versioned materials and reproduction tasks

The material inventory refers to the repository edition of 24 September 2026 at commit

4e67286ce72680a9e3a916a3fb7e97b1548b53f7.

The repository is https://github.com/AnXu-ITS/DeepSeek_Traveler_Behavior_Distillation_Research. It remains access-controlled. Its docs/REPRODUCIBILITY.md maps materials to reproduction tasks, and evidence/paper_20260924/ contains the artifact manifest, verification record and compact numerical evidence. The manuscript source package separately contains input definitions, adaptation settings, editable figures and tabulated summaries.

The authors confirm use of the official DeepSeek V4 Pro service throughout collection. The API documentation, checked on 25 September 2026, explicitly retains Pro service after 14 September and identifies the model as DeepSeek-V4-Pro-0813. This supersedes the routing assumption in the earlier editorial draft. The historical news page and the revised API documentation are not consistent on the announced routing plan; we use the updated service documentation and retained request records for attribution rather than infer that these experiments used Flash. The deduplicated

Table S38

Teacher targets, reference-model provenance and supporting materials. Frozen-target re-evaluation and live-API replication are distinct tasks.

Use	Coverage	Materials and interpretation
Controlled supervision	437 test states; repeat analysis covers 373 states and 350 pairs.	Prepared states, aggregate targets, splits, 12 selected neural fits, MNL-S records and evaluation code are included. Individual repeats are unavailable for the 64 mechanism states; the authors identify the service as official DeepSeek V4 Pro, while immutable per-response backend identity is not retained for every target.
Supply adaptation	336 accessibility states; 1,502 valid queries.	Adaptation specification and released SA-Student and predecessor checkpoints are included. Retained response records and training history support the adaptation provenance.
Contemporary API reference	24 respondents per city; ten tasks; three successful elicitations per state.	Configuration, prompt, campaign summary and aggregate comparisons are included. The returned model string is the alias <code>deepseek-v4-pro</code> ; participant-linked payloads are retained separately. The alias alone does not establish backend identity.

supervision archives retain completion creation times from 21 August to 27 August 2026 (UTC), while the final same-task reference completions were created on 21 September 2026 from 16:35:54 to 17:01:40 UTC. These are provider creation times, not selection seeds. The final campaign has 1,440 usage-bearing responses, all returning `deepseek-v4-pro`, and each recorded request specifies 16,384 maximum tokens. Its copied campaign configuration reports 8,192; the request-level fields take precedence for this batch. Early repeat records do not retain a response model field, and immutable backend fingerprints are incomplete. Frozen targets therefore remain the basis for exact target-based re-evaluation; service identity alone does not turn the later comparison into a causal estimate of compression error.

Controlled-model evaluation. The prepared benchmark is in `outputs/matched_response_v1/bundle/`, selected neural fits in `outputs/matched_response_v1/train/`, and modular fits in `outputs/revision_20260921/baselines/`. Local inference uses a synthetic example and a frozen checkpoint. The archived-checkpoint helper extracts the recorded source and checks its fingerprint before evaluation. Historical fingerprints depend on platform-specific path separators; portability changes should be versioned rather than treated as exact replay. The repository's retained verification record reports local inference and re-evaluation of two selected neural checkpoints, not a new full training campaign.

Survey analyses. The repository includes the English questionnaire instruments, complete aggregate contrast tables and analysis implementations. Reproducing respondent-level resampling or participant-linked Teacher comparisons requires the separately retained participant records and appropriate privacy-aware access. Aggregate-table inspection is distinct from re-estimation of participant-level intervals. These restrictions do not apply to the synthetic benchmark targets and released model weights.

Network execution. The repository includes the 140-run ledger, 70 paired summaries, experiment driver and event-analysis code. Fresh Helsinki execution requires the fixed population, prepared road/transit supply and Java/MATSim runtime; the full event archive, approximately 57.5 GB, is retained separately. The implementation is specified by `scripts/revision_20260921/helsinki_execution.py` and `src/traveler_distillation/matsim/adapters.py`. Dependency records are provided, although the historical stages do not share one fully locked environment.

The retained ledgers contain 1,000 people in each completed run and no zero-feasible-mass case. Across conditions, 2,006 outbound PT assignments and 940 car assignments fell back to walking; these are repeated traveler records, not distinct people. No recorded outbound or return walking route consisted only of its origin link. The general adapter can produce an origin-link-only walk after an unsuccessful walking search, so this tested behavior does not establish handling of disconnected networks. These ledger checks assess the completed runs; the repository expansion itself did

not rerun all simulations.

Requests for access should be directed to the corresponding author, subject to participant privacy and third-party terms. The included manifest records artifact hashes and the material scope, while the verification record distinguishes executed checks from retained historical outputs. This inventory supersedes earlier availability text that predates the repository expansion.